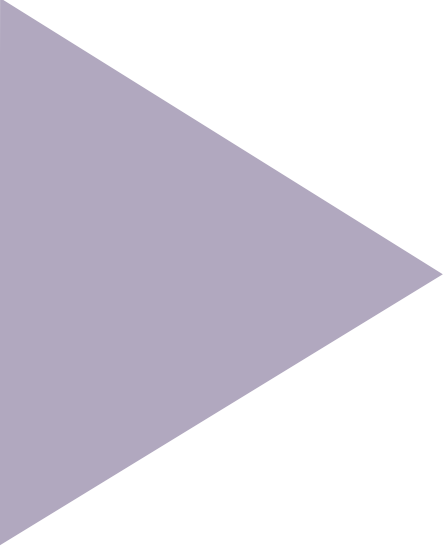




Tab 3: Risk Budgeting (Theory)

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<https://i2.wp.com/www.etfstrategy.co.uk/wp-content/uploads/2015/08/ThierryRoncalliLyxor.jpg?fit=300%2C200&ssl=1>

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Risk Budgeting

“... the performance of an allocation method may be strongly different depending on the choice of the constraints. In the end, we don't know if one allocation method is better than another, because of the portfolio method itself or because of the constraints. In this chapter, we present an allocation method requiring less discretionary inputs, namely the risk budgeting approach or the “risk parity” approach. The term risk parity is the technical term¹ used in the asset management industry.”

*Extract from Thierry Roncalli, Introduction to Risk Parity and Budgeting, CRC Press, 2014, pp. 71

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Risk Budgeting

2.1.1 Properties of a risk measure

Let $\mathcal{R}(x)$ be the risk measure of portfolio x . In this section, we define the different properties that should satisfy risk measure $\mathcal{R}(x)$ in order to be acceptable in terms of a risk allocation principle.

2.1.1.1 Coherency and convexity of risk measures

Following Artzner *et al.* (1999), $\mathcal{R}$ is said to be '*coherent*' if it satisfies the following properties:

*Extract from Thierry Roncalli, Introduction to Risk Parity and Budgeting, CRC Press, 2014, pp. 71

Risk Budgeting

1. Subadditivity

$$\mathcal{R}(x_1 + x_2) \leq \mathcal{R}(x_1) + \mathcal{R}(x_2)$$

The risk of two portfolios should be less than adding the risk of the two separate portfolios.

2. Homogeneity

$$\mathcal{R}(\lambda x) = \lambda \mathcal{R}(x) \quad \text{if } \lambda \geq 0$$

Leveraging or deleveraging of the portfolio increases or decreases the risk measure in the same magnitude.

3. Monotonicity

$$\text{if } x_1 \prec x_2, \text{ then } \mathcal{R}(x_1) \geq \mathcal{R}(x_2)$$

If portfolio x_2 has a better return than portfolio x_1 under all scenarios, risk measure $\mathcal{R}(x_1)$ should be higher than risk measure $\mathcal{R}(x_2)$.

4. Translation invariance

$$\text{if } m \in \mathbb{R}, \text{ then } \mathcal{R}(x + m) = \mathcal{R}(x) - m$$

Adding a cash position of amount m to the portfolio reduces the risk by m .

Risk Budgeting

By definition, the loss of the portfolio is $L(x) = -R(x)$ where $R(x)$ is the return of the portfolio. We consider then different risk measures:

- Volatility of the loss

$$\mathcal{R}(x) = \sigma(L(x)) = \sigma(x)$$

The volatility of the loss is the portfolio's volatility.

- Standard deviation-based risk measure

$$\mathcal{R}(x) = \text{SD}_c(x) = \mathbb{E}[L(x)] + c \cdot \sigma(L(x)) = -\mu(x) + c \cdot \sigma(x)$$

To obtain this measure, we scale the volatility by factor $c > 0$ and subtract the expected return of the portfolio.

Risk Budgeting

- Value-at-risk

$$\mathcal{R}(x) = \text{VaR}_\alpha(x) = \inf \{ \ell : \Pr \{ L(x) \leq \ell \} \geq \alpha \}$$

The value-at-risk is the α -quantile of the loss distribution $\mathbf{F}$ and we note it $\mathbf{F}^{-1}(\alpha)$.

- Expected shortfall

$$\mathcal{R}(x) = \text{ES}_\alpha(x) = \frac{1}{1-\alpha} \int_\alpha^1 \text{VaR}_u(x) \, du$$

The expected shortfall is the average of the VaRs at level α and higher (Acerbi and Tasche, 2002). We note that it is also equal to the expected loss given that the loss is beyond the value-at-risk:

$$\text{ES}_\alpha(x) = \mathbb{E}[L(x) \mid L(x) \geq \text{VaR}_\alpha(x)]$$

Risk Budgeting

Let us assume that the asset returns are normally distributed: $R \sim \mathcal{N}(\mu, \Sigma)$. We have $\mu(x) = x^\top \mu$ and $\sigma(x) = \sqrt{x^\top \Sigma x}$. It follows that the standard deviation-based risk measure is:

$$\text{SD}_c(x) = -x^\top \mu + c \cdot \sqrt{x^\top \Sigma x} \quad (2.1)$$

For the value-at-risk, we have $\Pr\{L(x) \leq \text{VaR}_\alpha(x)\} = \alpha$. Because $L(x) = -R(x)$ and $\Pr\{R(x) \geq -\text{VaR}_\alpha(x)\} = \alpha$, we deduce that:

$$\Pr\left\{\frac{R(x) - x^\top \mu}{\sqrt{x^\top \Sigma x}} \leq \frac{-\text{VaR}_\alpha(x) - x^\top \mu}{\sqrt{x^\top \Sigma x}}\right\} = 1 - \alpha$$

It follows that:

$$\frac{-\text{VaR}_\alpha(x) - x^\top \mu}{\sqrt{x^\top \Sigma x}} = \Phi^{-1}(1 - \alpha)$$

We finally obtain:

$$\text{VaR}_\alpha(x) = -x^\top \mu + \Phi^{-1}(\alpha) \sqrt{x^\top \Sigma x} \quad (2.2)$$

Risk Budgeting

This is a special case of the standard deviation-based risk measure with $c = \Phi^{-1}(\alpha)$. It implies that the value-at-risk is a coherent and convex risk measure if the asset returns are normally distributed. The expression of the expected shortfall is:

$$\text{ES}_\alpha(x) = \frac{1}{1 - \alpha} \int_{-\mu(x) + \sigma(x)\Phi^{-1}(\alpha)}^{\infty} \frac{u}{\sigma(x)\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{u + \mu(x)}{\sigma(x)}\right)^2\right) du$$

With the variable change $t = \sigma(x)^{-1}(u + \mu(x))$, we obtain:

$$\begin{aligned} \text{ES}_\alpha(x) &= \frac{1}{1 - \alpha} \int_{\Phi^{-1}(\alpha)}^{\infty} (-\mu(x) + \sigma(x)t) \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}t^2\right) dt \\ &= -\frac{\mu(x)}{1 - \alpha} [\Phi(t)]_{\Phi^{-1}(\alpha)}^{\infty} + \frac{\sigma(x)}{(1 - \alpha)\sqrt{2\pi}} \int_{\Phi^{-1}(\alpha)}^{\infty} t \exp\left(-\frac{1}{2}t^2\right) dt \\ &= -\mu(x) + \frac{\sigma(x)}{(1 - \alpha)\sqrt{2\pi}} \left[-\exp\left(-\frac{1}{2}t^2\right)\right]_{\Phi^{-1}(\alpha)}^{\infty} \\ &= -\mu(x) + \frac{\sigma(x)}{(1 - \alpha)\sqrt{2\pi}} \exp\left(-\frac{1}{2}[\Phi^{-1}(\alpha)]^2\right) \end{aligned}$$

Risk Budgeting

The expected shortfall of portfolio x is then:

$$\text{ES}_\alpha(x) = -x^\top \mu + \frac{\sqrt{x^\top \Sigma x}}{(1 - \alpha)} \phi(\Phi^{-1}(\alpha)) \quad (2.3)$$

Like the value-at-risk, it is a standard deviation-based risk measure with $c = \phi(\Phi^{-1}(\alpha)) / (1 - \alpha)$.

Example
Risk Budgeting
VaR and ES



Risk Budgeting

Example 6 We consider three stocks A , B and C . Their current prices are respectively 244, 135 and 315 dollars. We assume that their expected returns are equal to 50 bps, 30 bps and 20 bps on a daily basis, whereas their daily volatilities are 2%, 3% and 1%. The correlation of asset returns is given by the following matrix:

$$\rho = \begin{pmatrix} 1.00 & & \\ 0.50 & 1.00 & \\ 0.25 & 0.60 & 1.00 \end{pmatrix}$$

Risk Budgeting

We consider Portfolio #1 composed of two stocks A , one stock B and one stock C . The value of the portfolio is then equal to 938 dollars. We deduce that the weights x are (52.03%, 14.39%, 33.58%). Computation gives $\mu(x) = 37.0$ bps and $\sigma(x) = 1.476\%$. In Table 2.1, we report the values taken by the value-at-risk and the expected shortfall for different confidence levels α . Because these risk measures are computed using Formulas (2.2) and (2.3) with the portfolio weights, the values are expressed as a percentage. For example, the value-at-risk of the portfolio for $\alpha = 99\%$ is equal to²:

$$\text{VaR}_{99\%}(x) = -0.370\% + 2.326 \times 1.476\% = 3.06\%$$

For the expected shortfall, we obtain³:

$$\text{ES}_{99\%}(x) = -0.370\% + 2.667 \times 1.476\% = 3.56\%$$

Risk Budgeting

TABLE 2.1: Computation of risk measures $\text{VaR}_\alpha(x)$ and $\text{ES}_\alpha(x)$

Portfolio	$\mathcal{R}(x)$		α			
			90%	95%	99%	99.5%
#1	VaR	(in %)	1.52	2.06	3.06	3.43
		(in \$)	14.27	19.30	28.74	32.20
	ES	(in %)	2.22	2.67	3.56	3.90
		(in \$)	20.83	25.09	33.44	36.58
#2	VaR	(in %)	5.68	7.45	10.76	11.98
		(in \$)	14.94	19.59	28.31	31.50
	ES	(in %)	7.98	9.48	12.41	13.52
		(in \$)	21.00	24.94	32.64	35.54

Risk Budgeting

We can also express the risk measures in dollars. In this case, it suffices to replace the weights x by the nominal exposures in the previous formulas. In our example, these nominal exposures are (\$488, \$135, \$315). For $\alpha = 99\%$, the value-at-risk (resp. the expected shortfall) is equal to 28.74 dollars (resp. 33.44 dollars). Another way to compute the risk measure in dollars is to note that the loss expressed in nominal terms is:

$$\begin{aligned} L(x) &= P_{t+1}(x) - P_t(x) \\ &= P_t(x) \cdot (1 + R_{t+1}(x)) - P_t(x) \\ &= P_t(x) \cdot R_{t+1}(x) \end{aligned}$$

where $P_t(x)$ is the value of the portfolio at time t . By using the homogeneity property, we deduce that the risk measure in dollars is equal to the product of the risk measure in % by the current value of the portfolio. Thus we verify

Risk Budgeting

that $3.06\% \times 938 = \$28.74$. We now consider Portfolio #2 composed of two stocks A , one stock C and a short position in stock B . The results are also given in Table 2.1. We note that the second portfolio is more risky than the first portfolio in relative terms. However, they present a similar risk in dollars. The problem is that measuring the risk in percent is not appropriate for long-short portfolios. Suppose for example that the long position matches perfectly the short position in dollars. In this case, the value of the portfolio is zero and the relative risk measure is infinite. This is why it is better to reserve the computation of risk measures in percent to long-only portfolios.



Risk Budgeting

2.1.1.2 Euler allocation principle

Measuring risk is only the first step of portfolio risk management. It must be completed by a second step consisting in decomposing the risk portfolio into a sum of risk contributions by assets. This second step is also called *risk allocation* (Litterman, 1996). The concept of risk contribution is key in identifying concentrations and understanding the risk profile of the portfolio, and there are different methods for defining them. As illustrated by Denault (2001), some methods are more pertinent than others and the Euler principle is certainly the most used and accepted one.

Let Π be the P&L of the portfolio. We decompose it as the sum of the n asset P&L:

$$\Pi = \sum_{i=1}^n \Pi_i$$

Risk Budgeting

We note $\mathcal{R}(\Pi)$ the risk measure⁴ associated with the P&L. Let us consider the risk-adjusted performance measure (RAPM) defined by:

$$\text{RAPM}(\Pi) = \frac{\mathbb{E}[\Pi]}{\mathcal{R}(\Pi)}$$

Tasche (2008) considers the portfolio-related RAPM of the i^{th} asset defined by:

$$\text{RAPM}(\Pi_i | \Pi) = \frac{\mathbb{E}[\Pi_i]}{\mathcal{R}(\Pi_i | \Pi)}$$

Risk Budgeting

Tasche (2008) shows therefore that if there are risk contributions that are RAPM compatible in the sense of the two previous properties (2.4) and (2.5), then $\mathcal{R}(\Pi_i | \Pi)$ is uniquely determined as:

$$\mathcal{R}(\Pi_i | \Pi) = \left. \frac{d}{dh} \mathcal{R}(\Pi + h\Pi_i) \right|_{h=0} \quad (2.6)$$

and the risk measure is homogeneous of degree 1. In the case of a subadditive risk measure, one can also show that:

$$\mathcal{R}(\Pi_i | \Pi) \leq \mathcal{R}(\Pi_i) \quad (2.7)$$

This means that the risk contribution of asset i is always smaller than its stand-alone risk measure. The difference is related to the risk diversification.

Risk Budgeting

Let us return to risk measure $\mathcal{R}(x)$ defined in terms of weights. The previous framework implies that the risk contribution of asset i is uniquely defined as:

$$\mathcal{RC}_i = x_i \frac{\partial \mathcal{R}(x)}{\partial x_i} \quad (2.8)$$

and the risk measure satisfies the Euler decomposition:

$$\mathcal{R}(x) = \sum_{i=1}^n x_i \frac{\partial \mathcal{R}(x)}{\partial x_i} = \sum_{i=1}^n \mathcal{RC}_i \quad (2.9)$$

This relationship is also called the Euler allocation principle. It is the core of risk parity portfolios and is currently used intensively by practitioners.

Risk Budgeting

2.1.2 Risk contribution of portfolio assets

2.1.2.1 Computing the risk contributions

In the case of Gaussian asset returns, risk measures such as value-at-risk or expected shortfall are related to the volatility. Let us first consider the two-asset case. We have:

$$\sigma(x) = \sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}$$

It follows that the marginal risk of the first asset is:

$$\begin{aligned} \frac{\partial \sigma(x)}{\partial x_1} &= \frac{2x_1 \sigma_1^2 + 2x_2 \rho \sigma_1 \sigma_2}{2\sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}} \\ &= \frac{x_1 \sigma_1^2 + x_2 \rho \sigma_1 \sigma_2}{\sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}} \\ &= \frac{\text{cov}(R_1, R(x))}{\text{var}(R(x))} \end{aligned}$$

Risk Budgeting

We then deduce that the risk contribution of the first asset is:

$$\begin{aligned} \mathcal{RC}_1 &= x_1 \frac{\partial \sigma(x)}{\partial x_1} \\ &= \frac{x_1^2 \sigma_1^2 + x_1 x_2 \rho \sigma_1 \sigma_2}{\sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}} \end{aligned}$$

We verify that the sum of the two risk contributions is equal to the portfolio's volatility:

$$\begin{aligned} \mathcal{RC}_1 + \mathcal{RC}_2 &= \frac{x_1^2 \sigma_1^2 + x_1 x_2 \rho \sigma_1 \sigma_2}{\sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}} + \frac{x_2 x_1 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}{\sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}} \\ &= \frac{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}{\sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2}} \\ &= \sqrt{x_1^2 \sigma_1^2 + 2x_1 x_2 \rho \sigma_1 \sigma_2 + x_2^2 \sigma_2^2} \\ &= \sigma(x) \end{aligned}$$

Risk Budgeting

The previous formulas can be extended to the case $n > 2$. Because $\sigma(x) = \sqrt{x^\top \Sigma x}$, it follows that the vector of marginal volatilities is:

$$\begin{aligned}\frac{\partial \sigma(x)}{\partial x} &= \frac{1}{2} (x^\top \Sigma x)^{-1} (2 \Sigma x) \\ &= \frac{\Sigma x}{\sqrt{x^\top \Sigma x}}\end{aligned}$$

Risk Budgeting

The risk contribution of the i^{th} asset is then:

$$\mathcal{RC}_i = x_i \cdot \frac{(\Sigma x)_i}{\sqrt{x^\top \Sigma x}}$$

Like the two-asset case, we verify the full allocation property:

$$\begin{aligned}\sum_{i=1}^n \mathcal{RC}_i &= \sum_{i=1}^n x_i \cdot \frac{(\Sigma x)_i}{\sqrt{x^\top \Sigma x}} \\ &= x^\top \frac{\Sigma x}{\sqrt{x^\top \Sigma x}} \\ &= \sqrt{x^\top \Sigma x} \\ &= \sigma(x)\end{aligned}$$

Risk Budgeting

We can then deduce the risk contribution of the three risk measures $SD_c(x)$, $VaR_\alpha(x)$ and $ES_\alpha(x)$. For the standard deviation-based risk measure $SD_c(x)$, we have:

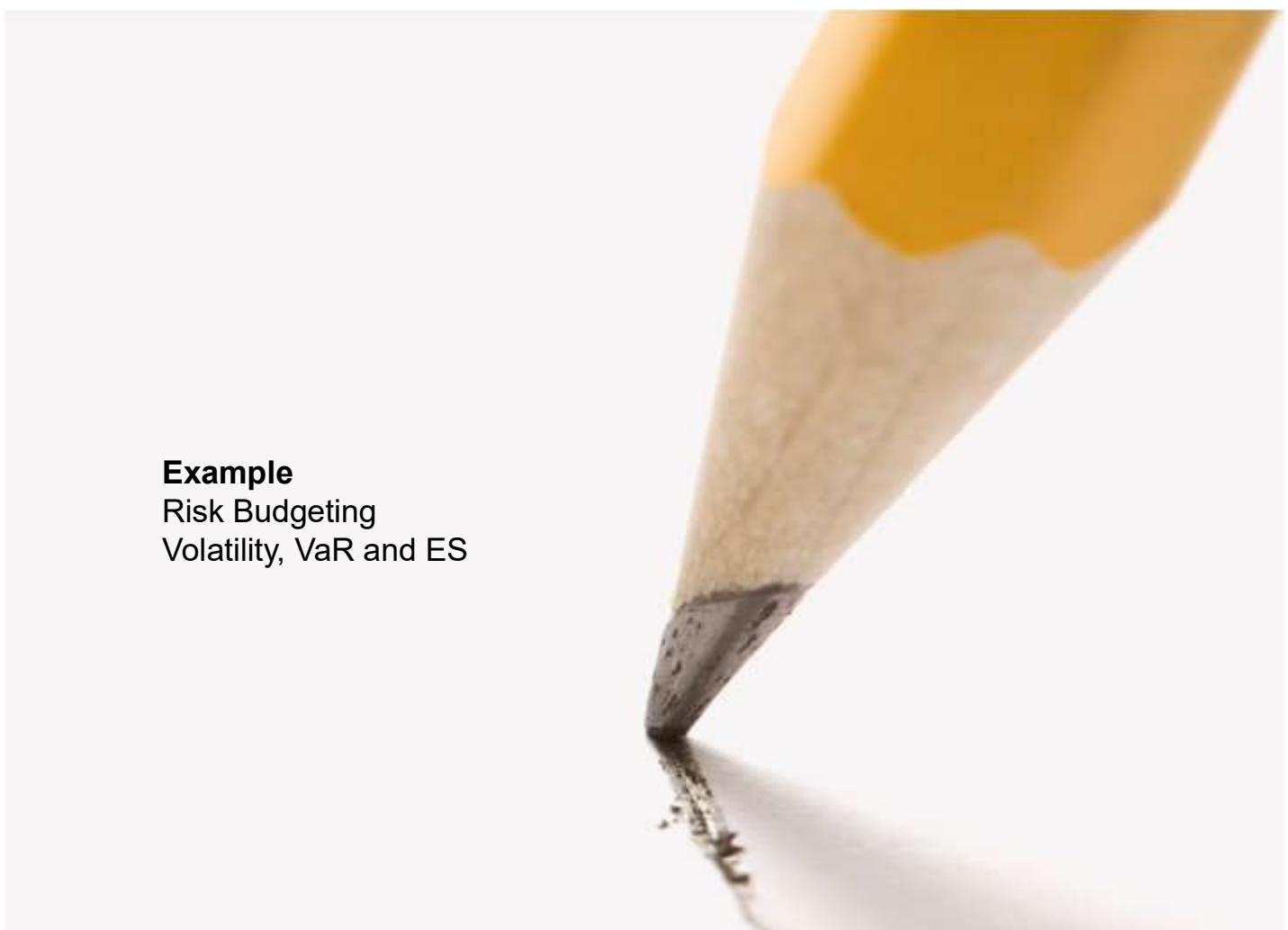
$$\mathcal{RC}_i = x_i \cdot \left(-\mu_i + c \frac{(\Sigma x)_i}{\sqrt{x^\top \Sigma x}} \right) \quad (2.10)$$

In the case of the value-at-risk, the risk contribution becomes:

$$\mathcal{RC}_i = x_i \cdot \left(-\mu_i + \Phi^{-1}(\alpha) \frac{(\Sigma x)_i}{\sqrt{x^\top \Sigma x}} \right) \quad (2.11)$$

whereas in the case of the expected shortfall, it is:

$$\mathcal{RC}_i = x_i \cdot \left(-\mu_i + \frac{(\Sigma x)_i}{(1 - \alpha) \sqrt{x^\top \Sigma x}} \phi(\Phi^{-1}(\alpha)) \right) \quad (2.12)$$



Example
Risk Budgeting
Volatility, VaR and ES

Risk Budgeting

Example 7 We consider three assets. We assume that their expected returns are equal to zero whereas their volatilities are equal to 30%, 20% and 15%. The correlation of asset returns is given by the following matrix:

$$\rho = \begin{pmatrix} 1.00 & & \\ 0.80 & 1.00 & \\ 0.50 & 0.30 & 1.00 \end{pmatrix}$$

Let us consider the portfolio (50%, 20%, 30%) in the case of Example 7. Using the relationship $\Sigma_{i,j} = \rho_{i,j}\sigma_i\sigma_j$, we deduce that the covariance matrix is⁶:

$$\Sigma = \begin{pmatrix} 9.00 & 4.80 & 2.25 \\ 4.80 & 4.00 & 0.90 \\ 2.25 & 0.90 & 2.25 \end{pmatrix} \times 10^{-2}$$

Risk Budgeting

It follows that the variance of the portfolio is:

$$\begin{aligned}\sigma^2(x) &= 0.50^2 \times 0.09 + 0.20^2 \times 0.04 + 0.30^2 \times 0.0225 + \\ &\quad 2 \times 0.50 \times 0.20 \times 0.0480 + 2 \times 0.50 \times 0.30 \times 0.0225 + \\ &\quad 2 \times 0.20 \times 0.30 \times 0.0090 \\ &= 4.3555\%\end{aligned}$$

The volatility is then $\sigma(x) = \sqrt{4.3555\%} = 20.8698\%$. The computation of the marginal volatilities gives:

$$\frac{\Sigma x}{\sqrt{x^\top \Sigma x}} = \frac{1}{20.8698\%} \begin{pmatrix} 6.1350\% \\ 3.4700\% \\ 1.9800\% \end{pmatrix} = \begin{pmatrix} 29.3965\% \\ 16.6269\% \\ 9.4874\% \end{pmatrix}$$

Risk Budgeting

Finally, we obtain the risk contributions by multiplying the weights by the marginal volatilities:

$$x \circ \frac{\Sigma x}{\sqrt{x^\top \Sigma x}} = \begin{pmatrix} 50\% \\ 20\% \\ 30\% \end{pmatrix} \circ \begin{pmatrix} 29.3965\% \\ 16.6269\% \\ 9.4874\% \end{pmatrix} = \begin{pmatrix} 14.6982\% \\ 3.3254\% \\ 2.8462\% \end{pmatrix}$$

We verify that the sum of risk contributions is equal to the volatility:

$$\sum_{i=1}^3 \mathcal{RC}_i = 14.6982\% + 3.3254\% + 2.8462\% = 20.8698\%$$

TABLE 2.2: Risk decomposition of the volatility

Asset	x_i	$\mathcal{MR}_i$	$\mathcal{RC}_i$	$\mathcal{RC}_i^*$
1	50.00	29.40	14.70	70.43
2	20.00	16.63	3.33	15.93
3	30.00	9.49	2.85	13.64
$\mathcal{R}(x)$	20.87			

Risk Budgeting

These different results⁷ are summarized in Table 2.2. For the value-at-risk or the expected shortfall, the marginal risks and the risk contributions are scaled by the corresponding value of c . We obtain results given in Table 2.3 and Table 2.4. We note that the first asset represents 50% of the weight, but about 70% of the portfolio's risk. Similarly, the third asset has a weight of 30%, but only a relative risk contribution of 13.64%. Contrary to the first asset, it generates less risk than its weight.

Risk Budgeting

TABLE 2.3: Risk decomposition of the value-at-risk

Asset	x_i	$\mathcal{MR}_i$	$\mathcal{RC}_i$	$\mathcal{RC}_i^*$
1	50.00	68.39	34.19	70.43
2	20.00	38.68	7.74	15.93
3	30.00	22.07	6.62	13.64
$\mathcal{R}(x)$				48.55

TABLE 2.4: Risk decomposition of the expected shortfall

Asset	x_i	$\mathcal{MR}_i$	$\mathcal{RC}_i$	$\mathcal{RC}_i^*$
1	50.00	78.35	39.17	70.43
2	20.00	44.31	8.86	15.93
3	30.00	25.29	7.59	13.64
$\mathcal{R}(x)$				55.62

Risk Budgeting

2.1.3 Application to non-normal risk measures

2.1.3.1 Non-normal value-at-risk and expected shortfall

In the previous section, we provided formulas for when asset returns are Gaussian. However, the previous expressions can be extended in the general case. For the value-at-risk, *Gourieroux et al. (2000)* show that the risk contribution is equal to⁹:

$$\begin{aligned}\mathcal{RC}_i &= \mathcal{R}(\Pi_i | \Pi) \\ &= -\mathbb{E}[\Pi_i | \Pi = -\text{VaR}_\alpha(\Pi)] \\ &= \mathbb{E}[L_i | L(x) = \text{VaR}_\alpha(L)]\end{aligned}\tag{2.15}$$

Risk Budgeting

Formula (2.15) is more general than Expression (2.11) obtained in the Gaussian case. Indeed, we can retrieve the latter if we assume that the returns are Gaussian. We recall that the portfolio return is $R(x) = \sum_{i=1}^n x_i R_i = x^\top R$. The portfolio loss is defined by $L(x) = -R(x)$. We deduce that:

$$\begin{aligned}\mathcal{RC}_i &= \mathbb{E}[-x_i R_i \mid -R(x) = \text{VaR}_\alpha(x)] \\ &= -x_i \cdot \mathbb{E}[R_i \mid R(x) = -\text{VaR}_\alpha(x)]\end{aligned}$$

Because $R(x)$ is a linear combination of R , the random vector $(R, R(x))$ is Gaussian and we have:

$$\begin{pmatrix} R \\ R(x) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu \\ x^\top \mu \end{pmatrix}, \begin{pmatrix} \Sigma & \Sigma x \\ x^\top \Sigma & x^\top \Sigma x \end{pmatrix}\right)$$

Risk Budgeting

We know that $\text{VaR}_\alpha(x) = -x^\top \mu + \Phi^{-1}(\alpha) \sqrt{x^\top \Sigma x}$. It follows that¹⁰:

$$\begin{aligned}\mathbb{E}[R \mid R(x) = -\text{VaR}_\alpha(x)] &= \mathbb{E}\left[R \mid R(x) = x^\top \mu - \Phi^{-1}(\alpha) \sqrt{x^\top \Sigma x}\right] \\ &= \mu + \Sigma x (x^\top \Sigma x)^{-1} \cdot \\ &\quad \left(x^\top \mu - \Phi^{-1}(\alpha) \sqrt{x^\top \Sigma x} - x^\top \mu\right) \\ &= \mu - \Phi^{-1}(\alpha) \Sigma x \frac{\sqrt{x^\top \Sigma x}}{(x^\top \Sigma x)^{-1}} \\ &= \mu - \Phi^{-1}(\alpha) \frac{\Sigma x}{\sqrt{x^\top \Sigma x}}\end{aligned}$$

We obtain finally the same expression as Equation (2.11):

$$\begin{aligned}\mathcal{RC}_i &= -x_i \cdot \left(\mu - \Phi^{-1}(\alpha) \frac{\Sigma x}{\sqrt{x^\top \Sigma x}}\right)_i \\ &= x_i \cdot \left(-\mu_i + \Phi^{-1}(\alpha) \frac{(\Sigma x)_i}{\sqrt{x^\top \Sigma x}}\right)\end{aligned}$$

Risk Budgeting

In the same way, Tasche (2002) shows that the general expression of the risk contributions for the expected shortfall is:

$$\begin{aligned}\mathcal{RC}_i &= \mathcal{R}(\Pi_i | \Pi) \\ &= -\mathbb{E}[\Pi_i | \Pi \leq -\text{VaR}_\alpha(\Pi)] \\ &= \mathbb{E}[L_i | L(x) \geq \text{VaR}_\alpha(L)]\end{aligned}\tag{2.16}$$

Using Bayes' theorem, it follows that:

$$\mathcal{RC}_i = \frac{\mathbb{E}[L_i \cdot \mathbf{1}\{L \geq \text{VaR}_\alpha(L)\}]}{1 - \alpha}$$

If we apply the previous formula to the Gaussian case, we obtain:

$$\mathcal{RC}_i = -\frac{x_i}{1 - \alpha} \mathbb{E}[R_i \cdot \mathbf{1}\{R(x) \leq -\text{VaR}_\alpha(L)\}]$$

We know that the random vector $(R_i, R(x))$ has a bivariate normal distribution with:

$$\begin{pmatrix} R_i \\ R(x) \end{pmatrix} \sim \mathcal{N}\left(\begin{pmatrix} \mu_i \\ x^\top \mu \end{pmatrix}, \begin{pmatrix} \Sigma_{i,i} & (\Sigma x)_i \\ (\Sigma x)_i & x^\top \Sigma x \end{pmatrix}\right)$$

Let $I = \mathbb{E}[R_i \cdot \mathbf{1}\{R(x) \leq -\text{VaR}_\alpha(L)\}]$. We note f the density function of the

Risk Budgeting

We finally obtain that:

$$\mathcal{RC}_i = x_i \cdot \left(-\mu_i + \frac{(\Sigma x)_i}{(1 - \alpha) \sqrt{x^\top \Sigma x}} \phi(\Phi^{-1}(\alpha)) \right)$$

This is the same expression as found previously.

When returns are not Gaussian, we use Equations (2.15) and (2.16) to compute the risk contributions. Because an analytical expression is generally difficult to find, we consider Monte Carlo simulations. Let m and $L_i^{(j)}$ be the number of simulations and the j^{th} simulated loss of the i^{th} asset. The loss of the portfolio for the simulation j is then equal to $L^{(j)} = \sum_{i=1}^n L_i^{(j)}$. We consider the order statistics¹⁴:

$$\min(L^{(j)}) = L^{(1:m)} \leq L^{(2:m)} \leq \dots \leq L^{(m:m)} = \max(L^{(j)})$$

Risk Budgeting

2.1.3.2 Historical value-at-risk

The historical value-at-risk consists in computing the quantile of the empirical distribution of losses using a set of historical scenarios, typically the last 260 trading days. It is also called the non-parametric approach, because it does not involve estimating parameters like in the Gaussian VaR. Let $\hat{\mathbf{F}}$ be the empirical distribution of loss portfolio $L(x)$. The historical VaR is then:

$$\text{VaR}_\alpha(L) = \hat{\mathbf{F}}^{-1}(\alpha)$$

To estimate this quantity, we can proceed as in the case of Monte Carlo simulations. In this case, m , $L_i^{(j)}$ and $L^{(j)}$ denote respectively the number of historical scenarios, the loss of the i^{th} asset for the j^{th} scenario and the portfolio loss for the j^{th} scenario. We have seen that the value-at-risk is then equal to the m_α order statistic: $\text{VaR}_\alpha(L) = L^{(m_\alpha:m)}$. If m_α is not an integer, we consider an interpolation between the two order statistics enclosing the confidence level α . We have:

$$\text{VaR}_\alpha(L) = L^{(m_\alpha:m)} + (m\alpha + 1 - m_\alpha) \left(L^{(m_\alpha+1:m)} - L^{(m_\alpha:m)} \right)$$

Risk Budgeting

where $m_\alpha = \lfloor m\alpha + 1 \rfloor$ is the integer part of $m\alpha + 1$. For example, if $\alpha = 99\%$ and $m = 250$, we obtain:

$$\begin{aligned} \text{VaR}_{99\%}(L) &= L^{(248:250)} + \frac{1}{2} \left(L^{(249:250)} - L^{(248:250)} \right) \\ &= - \left(\Pi^{(3:250)} + \frac{1}{2} \left(\Pi^{(2:250)} - \Pi^{(3:250)} \right) \right) \\ &= - \left(\frac{1}{2} \Pi^{(2:250)} + \frac{1}{2} \Pi^{(3:250)} \right) \end{aligned}$$

Computing the 99% value-at-risk with 250 historical scenarios is then equal to the opposite of the average of the second and third smallest P&Ls.

The risk contribution estimation faces the same problem as in the Monte Carlo case. If asset returns are assumed to be elliptically distributed, Carroll *et al.* (2001) shows that¹⁸:

$$\mathcal{RC}_i = \mathbb{E}[L_i] + \frac{\text{cov}(L, L_i)}{\sigma^2(L)} (\text{VaR}_\alpha(L) - \mathbb{E}[L]) \quad (2.17)$$

Risk Budgeting

Estimating the risk contributions with the historical value-at-risk is then straightforward. It suffices to apply Formula (2.17) by replacing the statistical moments by their sample statistics:

$$\mathcal{RC}_i = \bar{L}_i + \frac{\sum_{j=1}^m (L^{(j)} - \bar{L}) (L_i^{(j)} - \bar{L}_i)}{\sum_{j=1}^m (L^{(j)} - \bar{L})^2} (\text{VaR}_\alpha(L) - \bar{L})$$

where $\bar{L}_i = m^{-1} \sum_{j=1}^m L_i^{(j)}$ and $\bar{L} = m^{-1} \sum_{j=1}^m L^{(j)}$.

Result (2.17) can be viewed as the estimation of the conditional expectation $\mathbb{E}[L_i | L = \text{VaR}_\alpha(L)]$ in a linear regression framework:

$$L_i = \beta L + \varepsilon_i$$

Because the least squared estimator is $\hat{\beta} = \text{cov}(L, L_i) / \sigma^2(L)$, we deduce that:

$$\begin{aligned} \mathbb{E}[L_i | L = \text{VaR}_\alpha(L)] &= \hat{\beta} \text{VaR}_\alpha(L) + \mathbb{E}[\varepsilon_i] \\ &= \hat{\beta} \text{VaR}_\alpha(L) + (\mathbb{E}[L_i] - \hat{\beta} \mathbb{E}[L]) \\ &= \mathbb{E}[L_i] + \hat{\beta} (\text{VaR}_\alpha(L) - \mathbb{E}[L]) \end{aligned}$$

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Knowledge | Skills | Conduct

Risk Budgeting

Professionals prefer to use historical value-at-risk instead of Gaussian value-at-risk because it depends (implicitly) on all the moments of the loss distribution. Another approach consists in correcting the Gaussian value-at-risk by taking into account the third and fourth moments. Let $\mu_r = \mathbb{E}[(L - \mu(L))^r]$ be the centered r -order moment of the portfolio loss²⁰. The skewness $\gamma_1 = \mu_3 / \mu_2^{3/2}$ is the measure of the asymmetry of the loss distribution²¹. To characterize whether the loss is peaked or flat relative to the normal distribution, we consider the excess kurtosis²² $\gamma_2 = \mu_4 / \mu_2^2 - 3$. Using the Cornish-Fisher expansion of the distribution, Zangari (1996) proposes to estimate the value-at-risk in the following way:

$$\text{VaR}_\alpha(L) = -x^\top \mu + z \cdot \sqrt{x^\top \Sigma x}$$

where:

$$z = z_\alpha + \frac{1}{6} (z_\alpha^2 - 1) \gamma_1 + \frac{1}{24} (z_\alpha^3 - 3z_\alpha) \gamma_2 - \frac{1}{36} (2z_\alpha^3 - 5z_\alpha) \gamma_1^2$$

with $z_\alpha = \Phi^{-1}(\alpha)$. This is the same formula as the one used for the Gaussian value-at-risk but with another scaling parameter²³. We deduce that the expression of the risk contribution is:

$$\mathcal{RC}_i = x_i \cdot \left(-\mu_i + z \cdot \frac{(\Sigma x)_i}{\sqrt{x^\top \Sigma x}} + \partial_{x_i} z \cdot \sqrt{x^\top \Sigma x} \right)$$



Risk Budgeting

2.2.1 Definition of a risk budgeting portfolio

We consider a set of given risk budgets $\{B_1, \dots, B_n\}$. Here, B_i is an amount of risk measured in dollars. We note $\mathcal{RC}_i(x_1, \dots, x_n)$ the risk contribution of asset i with respect to portfolio $x = (x_1, \dots, x_n)$. The risk budgeting portfolio is then defined by the following constraints:

$$\begin{cases} \mathcal{RC}_1(x_1, \dots, x_n) = B_1 \\ \vdots \\ \mathcal{RC}_i(x_1, \dots, x_n) = B_i \\ \vdots \\ \mathcal{RC}_n(x_1, \dots, x_n) = B_n \end{cases} \quad (2.19)$$

Risk Budgeting

It is therefore the portfolio x such that the risk contributions match the risk budgets. We note that there are two main differences between a risk budgeting portfolio and an optimized portfolio:

1. A risk budgeting portfolio is not based on the maximization of a utility function.
2. A risk budgeting portfolio does not depend (explicitly) on the expected performance of the portfolio.

Contrary to the Markowitz approach, we only consider the risk dimension. The main idea is that the performance dimension is too complicated to encompass, because the forecasting step is generally not robust. This is why we focus on the patterns of the portfolio risk.



Example
Risk Budgeting
Budegting contstraints

Risk Budgeting

Example 11 We consider three asset classes. Their expected returns are equal to 10%, 5% and 9% whereas their volatilities are equal to 30%, 20% and 15%. The correlation of asset returns is given by the following matrix:

$$\rho = \begin{pmatrix} 1.00 & & \\ 0.80 & 1.00 & \\ 0.50 & 0.30 & 1.00 \end{pmatrix}$$

We consider the example above. The risk measure of the investor is the normal expected shortfall and his risk limit is an annual loss of \$500 000. Additionally to this global risk limit, the investor has other risk limits fixed by asset classes: \$250 000 for the first asset class, \$100 000 for the second asset class and \$150 000 for the third asset class. We face here a risk budgeting problem with $B_1 = \$250\,000$, $B_2 = \$100\,000$ and $B_3 = \$150\,000$. If we consider the confidence levels 95% and 99%, we obtain the results given in Tables 2.9 and 2.10. For each portfolio, we have indicated the nominal exposure x_i , the weight w_i , the marginal risk $\mathcal{MR}_i$, the nominal risk contribution $\mathcal{RC}_i$ and the normalized risk contribution $\mathcal{RC}_i^*$. If α is set to 95%, the total exposure is equal to 1 894 142 and the composition of the portfolio is (28.21%, 19.68%, 52.11%). We note that these figures change if α is equal to 99%. The reason for this is that the expected shortfall is not homogeneous with respect to α because of the expected loss term.

Risk Budgeting

TABLE 2.9: Risk budgeting portfolio when the risk measure is the expected shortfall ($\alpha = 95\%$)

Asset	x_i	w_i	$\mathcal{MR}_i$	$\mathcal{RC}_i$	$\mathcal{RC}_i^*$
1	534 430	28.21%	46.78%	250 000	50.00%
2	372 705	19.68%	26.83%	100 000	20.00%
3	987 007	52.11%	15.20%	150 000	30.00%
sum	1 894 142			500 000	

TABLE 2.10: Risk budgeting portfolio when the risk measure is the expected shortfall ($\alpha = 99\%$)

Asset	x_i	w_i	$\mathcal{MR}_i$	$\mathcal{RC}_i$	$\mathcal{RC}_i^*$
1	391 926	29.00%	63.79%	250 000	50.00%
2	273 737	20.26%	36.53%	100 000	20.00%
3	685 779	50.74%	21.87%	150 000	30.00%
sum	1 351 441			500 000	

Risk Budgeting

It is also easy to show that the solution of the system (2.19) with rescaled risk budgets $b_i = cB_i$ is $y = cx$. Bruder and Roncalli (2012) propose to simplify the problem by using weights instead of nominal exposures²⁵ and defining the problem with the relative risk budgets b_i expressed in %. These authors note also that risk budgeting techniques are used to build diversified portfolios. Specifying that some assets have a negative risk contribution implies that the risk is highly concentrated in the other assets of the portfolio. Generally, we prefer to obtain a long-only portfolio meaning that all the weights are positive. This is why we define a proper RB portfolio by the following non-linear system:

$$\begin{cases} \mathcal{RC}_i(x) = b_i \mathcal{R}(x) \\ b_i \geq 0 \\ x_i \geq 0 \\ \sum_{i=1}^n b_i = 1 \\ \sum_{i=1}^n x_i = 1 \end{cases} \quad (2.20)$$

Risk Budgeting

2.2.1.2 Solving the non-linear system of risk budgeting constraints

With the exception of some trivial cases, it is not possible to find an analytical solution to Problem (2.21). However, we can always find a numerical solution. The first approach is to consider Broyden or SQRF algorithms to solve the non-linear system (2.21), but they do not always converge. This is why it is better to transform the non-linear system into an optimization problem:

$$\begin{aligned} x^* &= \arg \min f(x; b) \\ \text{u.c. } & \mathbf{1}^\top x = 1 \quad \text{and} \quad \mathbf{0} \leq x \leq \mathbf{1} \end{aligned} \quad (2.22)$$

For example, we may specify the function $f(x; b)$ as follows:

$$f(x; b) = \sum_{i=1}^n (x_i \cdot \partial_{x_i} \mathcal{R}(x) - b_i \mathcal{R}(x))^2$$

Risk Budgeting

If x^* is the solution of (2.22) and if $f(x^*; b) = 0$, it is obvious that x^* is also the solution of (2.21). We note that there are other appropriate functions $f(x; b)$ to define the RB portfolio, for example:

$$f(x; b) = \sum_{i=1}^n \sum_{j=1}^n \left(\frac{x_i \cdot \partial_{x_i} \mathcal{R}(x)}{b_i} - \frac{x_j \cdot \partial_{x_j} \mathcal{R}(x)}{b_j} \right)^2$$

Remark 28 To solve Problem (2.22), we consider the sequential quadratic programming (or SQP) algorithm²⁷ presented in Appendix A.1.3. Computing the analytical Jacobian and Hessian of the function $f(x; b)$ helps reduce computational times, especially when the number of assets is large.

Example

Risk Budgeting

Budgeting constraints SQP



Risk Budgeting

2.3 Special case: the ERC portfolio

The ERC portfolio is a RB portfolio such that the risk budgets are the same for the different assets:

$$b_i = \frac{1}{n}$$

It can be viewed as a neutral portfolio when the manager has no views on the risk budgets. As a result, it has the same properties of the RB portfolio presented before, but it also has some specific properties which have been studied by Maillard *et al.* (2010) in the case of the volatility risk measure.

Risk Budgeting

2.3.1 The two-asset case ($n = 2$)

We note by σ_i the volatility of the i^{th} asset, ρ the correlation and $x = (w, 1 - w)$ the portfolio weights. The marginal risk contributions are:

$$\frac{\partial \sigma(x)}{\partial x} = \frac{\Sigma x}{\sqrt{x^\top \Sigma x}} = \frac{1}{\sigma(x)} \begin{pmatrix} w\sigma_1^2 + (1-w)\rho\sigma_1\sigma_2 \\ (1-w)\sigma_2^2 + w\rho\sigma_1\sigma_2 \end{pmatrix}$$

The ERC portfolio satisfies then:

$$w \cdot \frac{(w\sigma_1^2 + (1-w)\rho\sigma_1\sigma_2)}{\sigma(x)} = (1-w) \cdot \frac{((1-w)\sigma_2^2 + w\rho\sigma_1\sigma_2)}{\sigma(x)}$$

We deduce that:

$$w^2\sigma_1^2 = (1-w)^2\sigma_2^2$$

Because $0 \leq w \leq 1$, the unique solution is:

$$\begin{aligned} w^* &= \frac{1}{\sigma_1} \bigg/ \left(\frac{1}{\sigma_1} + \frac{1}{\sigma_2} \right) \\ &= \frac{\sigma_2}{\sigma_1 + \sigma_2} \end{aligned} \tag{2.33}$$

The weights of the ERC portfolio are inversely proportional to the volatilities. Curiously, they do not depend on the correlation ρ .

Risk Budgeting

Let us compare this portfolio with two other heuristic approaches, which are the equally weighted (or EW) portfolio and the minimum variance (or MV) portfolio. We use the parameterization $\sigma_1 = \sigma$ and $\sigma_2 = k\sigma$ with $k \geq 0$. The ERC portfolio volatility is:

$$\sigma(x_{\text{erc}}) = \sqrt{2(1+\rho)} \frac{k}{1+k} \sigma$$

In comparison, the volatility of an equally weighted portfolio is:

$$\sigma(x_{\text{ew}}) = \frac{1}{2} \sigma \sqrt{1+k^2+2\rho k}$$

Using these expressions, we can check that $\sigma(x_{\text{ew}}) \geq \sigma(x_{\text{erc}})$. The equality is obtained for $k=1$, that is when $x_{\text{erc}} = x_{\text{ew}}$. We recall that, in the two-asset case, the minimum variance portfolio is:

$$x_{\text{mv}} = \frac{1}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2} \begin{pmatrix} \sigma_2^2 - \rho\sigma_1\sigma_2 \\ \sigma_1^2 - \rho\sigma_1\sigma_2 \end{pmatrix}$$

Risk Budgeting

Therefore, its volatility is:

$$\sigma(x_{\text{mv}}) = k\sigma \sqrt{\frac{1-\rho^2}{1+k^2-2\rho k}}$$

Maillard *et al.* (2010) deduce finally that the volatility of the ERC portfolio is located between the volatility of the minimum variance portfolio and the equally weighted portfolio:

Example
Risk Budgeting
Equal Risk Contribution (ERC)
N=2

Risk Budgeting

Example 16 We consider two assets with $\sigma_1 = 20\%$ and $\sigma_2 = 30\%$. The correlation between the asset returns is set to 50%.

In Table 2.20, we have reported the composition of the EW, ERC and MV portfolios. We have also indicated the risk contributions of each portfolio. We note that $\sigma_{\text{ew}} = 21.79\%$, $\sigma_{\text{erc}} = 20.78\%$ and $\sigma_{\text{mv}} = 19.64\%$, meaning that the inequality $\sigma_{\text{mv}} \leq \sigma_{\text{erc}} \leq \sigma_{\text{ew}}$ holds.

TABLE 2.20: Risk contributions of EW, ERC and MV portfolios

Portfolio	Asset	x_i	$\mathcal{MR}_i$	$\mathcal{RC}_i$	$\mathcal{RC}_i^*$
EW	1	50.00	16.06	8.03	36.84
	2	50.00	27.53	13.76	63.16
ERC	1	60.00	17.32	10.39	50.00
	2	40.00	25.98	10.39	50.00
MV	1	85.71	19.64	16.83	85.71
	2	14.29	19.64	2.81	14.29

Risk Budgeting

2.3.2 The general case ($n > 2$)

As in the case of the RB portfolio, it is not possible to find an analytical solution when $n > 2$. However, we have some closed-form solutions in some specific cases, for instance when the correlation or the volatility is the same for the different assets of the investment universe.

If we assume a constant correlation matrix with $\rho_{i,j} = \rho$ for all i, j , the expression of the risk contribution is given by Equation (2.24). It follows that the weights of the ERC portfolio satisfy the following relationship:

$$x_i \sigma_i \left((1 - \rho) x_i \sigma_i + \rho \sum_{k=1}^n x_k \sigma_k \right) = x_j \sigma_j \left((1 - \rho) x_j \sigma_j + \rho \sum_{k=1}^n x_k \sigma_k \right)$$

It follows that $x_i \sigma_i = x_j \sigma_j$. Because $\sum_{i=1}^n x_i = 1$, we deduce that:

$$x_i = \frac{\sigma_i^{-1}}{\sum_{j=1}^n \sigma_j^{-1}} \quad (2.34)$$

Risk Budgeting

Like in the two-asset case, the weight allocated to asset i is inversely proportional to its volatility and does not depend on the value of the correlation. Moreover, we can show that this solution is connected to the minimum variance portfolio. Let $\Sigma = \sigma \sigma^\top \circ C_n(\rho)$ be the covariance matrix with $C_n(\rho)$ the constant correlation matrix. We have $\Sigma^{-1} = \Gamma \circ C_n^{-1}(\rho)$ with $\Gamma_{i,j} = \sigma_i^{-1} \sigma_j^{-1}$ and:

$$C_n^{-1}(\rho) = \frac{\rho \mathbf{1} \mathbf{1}^\top - ((n-1)\rho + 1) I_n}{(n-1)\rho^2 - (n-2)\rho - 1}$$

We saw that the minimum variance portfolio is $x = (\Sigma^{-1} \mathbf{1}) / (\mathbf{1}^\top \Sigma^{-1} \mathbf{1})$. We deduce that the expression of the MV weights are:

$$x_i = \frac{-((n-1)\rho + 1) \sigma_i^{-2} + \rho \sum_{j=1}^n (\sigma_i \sigma_j)^{-1}}{\sum_{k=1}^n \left(-((n-1)\rho + 1) \sigma_k^{-2} + \rho \sum_{j=1}^n (\sigma_k \sigma_j)^{-1} \right)}$$

We know that the lower bound of $C_n(\rho)$ is achieved for $\rho = -(n-1)^{-1}$. In

Risk Budgeting

this case, the solution becomes:

$$x_i = \frac{\sum_{j=1}^n (\sigma_i \sigma_j)^{-1}}{\sum_{k=1}^n \sum_{j=1}^n (\sigma_k \sigma_j)^{-1}} = \frac{\sigma_i^{-1}}{\sum_{k=1}^n \sigma_k^{-1}}$$

The ERC portfolio is then equal to the MV portfolio when the correlation is at its lowest possible value.

If we assume now that all volatilities are equal, i.e. $\sigma_i = \sigma$ for all i , the risk contribution becomes:

$$\mathcal{RC}_i = \frac{(\sum_{k=1}^n x_i x_k \rho_{i,k}) \sigma^2}{\sigma(x)}$$

Risk Budgeting

The ERC portfolio verifies then:

$$x_i \left(\sum_{k=1}^n x_k \rho_{i,k} \right) = x_j \left(\sum_{k=1}^n x_k \rho_{j,k} \right)$$

We deduce that:

$$x_i = \frac{(\sum_{k=1}^n x_k \rho_{i,k})^{-1}}{\sum_{j=1}^n (\sum_{k=1}^n x_k \rho_{j,k})^{-1}} \quad (2.35)$$

The weight of asset i is inversely proportional to the weighted average of correlations of asset i . Contrary to the previous case, this solution is endogenous since x_i is a function of itself directly. The same issue of endogeneity naturally arises in the general case where both the volatilities and the correlations differ. If we replace b_i by $1/n$ in Equation (2.29), we deduce that:

$$x_i = \frac{\beta_i^{-1}}{\sum_{j=1}^n \beta_j^{-1}}$$

Because $\sum_{i=1}^n x_i \beta_i = 1$, we finally obtain:

$$x_i = \frac{1}{n \beta_i} \quad (2.36)$$



Example
Risk Budgeting
Equal Risk Contribution (ERC)
N=4

Risk Budgeting

Example 17 We consider an investment universe of four assets with $\sigma_1 = 15\%$, $\sigma_2 = 20\%$, $\sigma_3 = 30\%$ and $\sigma_4 = 10\%$. The correlation of asset returns is given by the following matrix:

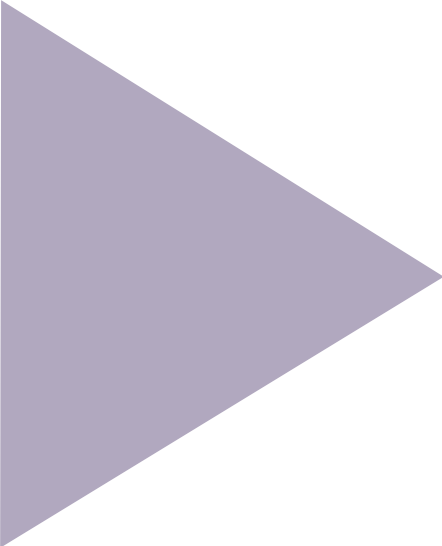
$$\rho = \begin{pmatrix} 1.00 & & & \\ 0.50 & 1.00 & & \\ 0.00 & 0.20 & 1.00 & \\ -0.10 & 0.40 & 0.70 & 1.00 \end{pmatrix}$$

The composition of the ERC is reported in Table 2.21. We verify that the weight of asset i is inversely proportional to its beta. Knowing the volatility, we can also deduce the risk contribution of each asset because we have $\mathcal{RC}_i = \sigma(x)/n$.

Risk Budgeting

TABLE 2.21: Composition of the ERC portfolio

Asset	x_i	$\mathcal{MR}_i$	β_i	$\mathcal{RC}_i$	$\mathcal{RC}_i^*$
1	31.34	8.52	0.80	2.67	25.00
2	17.49	15.27	1.43	2.67	25.00
3	13.05	20.46	1.92	2.67	25.00
4	38.12	7.00	0.66	2.67	25.00
Volatility				10.68	



The Naïve Risk Parity Strategy

Naïve Risk Parity

The simplest and most rudimental form of risk parity strategy is commonly referred to as naïve risk parity (Lee 2011; Maillard et al. 2010). This denomination can be explained by the simplified way used to identify the weight to allocate to each asset class. As a matter of fact, recommended weights are obtained without considering correlation information among asset classes and this makes naïve risk parity computationally very attractive.

In detail, the rule it follows for computing portfolio weights suggests an exposure for each asset class proportional to the inverse of its standard deviation; it is then normalized to guarantee the weights sum to 1. Accordingly, the analytical expression for asset class weight w_i is written as:

$$w_i = \frac{\sigma_i^{-1}}{\sum_{j=1}^N \sigma_j^{-1}} = \frac{\frac{1}{\sigma_i}}{\sum_{j=1}^N \frac{1}{\sigma_j}} \quad (3.19)$$

As a consequence of the Eq. (3.19), the higher (lower) is the volatility of asset class i , the lower (higher) is its weight in the portfolio. The rule used to implement the strategy motivates the alternative denomination of naïve risk parity as inverse volatility strategy.

Extracts from Maria Debora Braga, Risk-Based Approaches to Asset Allocation: Concepts and Practical Applications, Springer 2016

Example

Risk Budgeting

Equal Risk Contribution (ERC)

N=2



Naïve Risk Parity

In spite of its attractive computational features, it seems unlikely that naïve risk parity provides a true homogeneity in asset class contributions to the portfolio standard deviation. Indeed, there are just two cases where the naïve risk parity portfolio is consistent and equivalent with the optimal risk parity portfolio, that is it provides a perfect balance in terms of risk contributions:

- in the case of a bivariate investment universe, thus in case the portfolio constituents are just two;
- in the case of uniform correlations for every possible couples of asset classes.

It is intuitive that under these conditions, ignoring correlations has no impact.

In order to easily understand that equal risk contributions are obtained in the two-asset class case without taking into account correlation, let us look at the example illustrated in Sect. 3.1. In this case, the unique solution the naïve risk parity would propose is given by:

$$\begin{bmatrix} W_{JPM\ Emu\ Gov.\ All\ Mats} \\ W_{MSCI\ World} \end{bmatrix} = \begin{bmatrix} \frac{1/4.02\%}{1/4.02\% + 1/12.47\%} \\ \frac{1/12.47\%}{1/4.02\% + 1/12.47\%} \end{bmatrix} = \begin{bmatrix} 75.6\% \\ 24.4\% \end{bmatrix} \quad (3.20)$$

Naïve Risk Parity

With this structure, the portfolio risk goes from 5.27 to 4.033 %. Table 3.3 summarizes the main figures concerning risk decomposition.

Table 3.3 Risk decomposition for naïve risk parity portfolio in a two-asset class case

	JPM Emu Gov. All Mats	MSCI World
Naïve risk parity portfolio weights (%)	75.6	24.4
Marginal risk	0.026666	0.082717
Risk contribution (%)	2.016	2.016
Percentage risk contribution (%)	50	50

Naïve Risk Parity

These results show that for the two-asset class case the naïve risk parity solution for portfolio allocations to A and B can be written as:

$$\begin{bmatrix} w_A \\ w_B \end{bmatrix} = \begin{bmatrix} \frac{\sigma_A^{-1}}{\sigma_A^{-1} + \sigma_B^{-1}} \\ \frac{\sigma_B^{-1}}{\sigma_A^{-1} + \sigma_B^{-1}} \end{bmatrix} = \begin{bmatrix} \frac{1/\sigma_A}{1/\sigma_A + 1/\sigma_B} \\ \frac{1/\sigma_B}{1/\sigma_A + 1/\sigma_B} \end{bmatrix} \quad (3.21)$$

By presenting w_A simply as w and w_B as $(1-w)$, the allocations given in (3.21) are the only ones that satisfy the condition that the two risk contributions written in (3.22) are equal:

$$\frac{1}{\sigma_P} \left\{ \begin{aligned} &\sigma_A^2 w^2 + w(1-w)\sigma_A\sigma_B\rho_{A,B} \\ &\sigma_B^2 (1-w)^2 + w(1-w)\sigma_A\sigma_B\rho_{A,B} \end{aligned} \right. \quad (3.22)$$

Naïve Risk Parity

By illustrating the case of constant correlations, which means $\rho_{ij} = \rho \forall i, j$, we note that the marginal risk of asset class i becomes:

$$MR_{\sigma_i} = \frac{\sigma_i^2 w_i + \rho \sum_{j \neq i}^N w_j \sigma_i \sigma_j}{\sigma_P} \quad (3.23)$$

Then we get the total risk contribution from asset class i just multiplying (3.23) by w_i :

$$TRC_{\sigma_i} = \frac{\sigma_i^2 w_i^2 + \rho \sum_{j \neq i}^N w_j w_i \sigma_i \sigma_j}{\sigma_P} \quad (3.24)$$

We can rearrange (3.24) and obtain the following expression:

$$TRC_{\sigma_i} = \frac{\sigma_i w_i \left((1-\rho)\sigma_i w_i + \rho \sum_{j \neq i}^N w_j \sigma_j \right)}{\sigma_P} \quad (3.25)$$

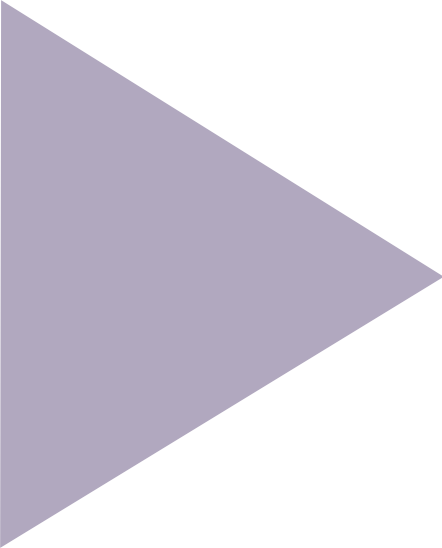
Naïve Risk Parity

Considering the relationship that should exist among risk contributions:

$$\frac{\sigma_i w_i ((1 - \rho) \sigma_i w_i + \rho \sum_{h=1}^N w_h \sigma_h)}{\sigma_P} = \frac{\sigma_j w_j ((1 - \rho) \sigma_j w_j + \rho \sum_{h=1}^N w_h \sigma_h)}{\sigma_P} \quad (3.26)$$

it is easy to deduce that the fact it is verified depends on the equality between $\sigma_i w_i$ and $\sigma_j w_j$. Recalling the existence of the budget constraint, it is necessary that the generic weight w_i is inversely related to its volatility.

At this point, in order to draw a conclusion about the risk-based strategy discussed in this section, we can say that, even if naïve risk parity is not a sound and authentic risk parity strategy (except in two scenarios), it leads to the intuition that the application of this rudimental strategy will provide solutions similar to those originating from the optimal risk parity we illustrate in Sect. 3.4 when correlations tend to show convergence to common, or rather, strongly similar values.



The Equally-weighted Approach

The Equally-weighted Approach

The most straightforward risk-based strategy to illustrate and to put into practice as well is represented by the equally-weighted approach. According to it, the portfolio allocations have only to be driven by the number of asset classes in the investment universe. This is obvious because the asset allocation problem is solved using the so called “1 over N” rule which prescribes that:

$$w_i = w_j = \frac{1}{N} \quad \forall i, j \quad (4.1)$$

Hence, when the equally-weighted approach is used, all asset classes are given an identical and static weight.¹ We can draw a sort of parallelism from this point between the equally-weighted strategy and the optimal risk parity strategy: the first

Extracts from Maria Debora Braga, Risk-Based Approaches to Asset Allocation: Concepts and Practical Applications, Springer 2016

The Equally-weighted Approach

one recommends the less concentrated portfolio in terms of weights, the second one suggests the less concentrated portfolio in terms of risk. Said differently, the former applies the “1 over N” rule to asset allocation while the latter applies the same concept to risk allocation.

Looking at (4.1), we can immediately see that the equally-weighted approach requires no investigation of the distribution of asset class returns. The portfolio allocation is determined regardless of any statistical estimate for returns, risks and correlations. Obviously, it involves no parameter estimate because its implementation does not entail an optimization problem to solve. Therefore, we cannot identify an objective function to minimize or maximize for the case of equally-weighted strategy.

Given this evidence, it is extremely difficult to assume a condition of ex-ante optimality, in a mean-variance framework, for the equally-weighted portfolio. It is certain that very demanding and improbable conditions would be required in order to assert that it is ex-ante an efficient portfolio according to Markowitz: equal expected returns, equal volatilities and uniform correlations among asset classes. In ex-post or empirical analysis we find contrasting results on this strategy: in a very extensive study, based on several asset allocation models and different datasets, DeMiguel et al. (2009) conclude that none of the theoretically more robust asset allocation models was consistently better out-of sample than the “1 over N” rule, but an opposite conclusion comes from Kritzman et al. (2010).

The Equally-weighted Approach

Apart from the discussion regarding the optimality ex-ante versus ex-post of the equally-weighted strategy, the considerations expressed before may also have another consequence. More precisely, they can stimulate some hesitation or suspicion about the inclusion of the equally-weighted approach among the risk-based strategies given that the estimates for the statistical second moments of asset classes too are not taken into consideration when the strategy is implemented. In order to provide an explanation for such inclusion, it is both useful and interesting to develop an idea about what investors try to achieve when they use the equally-weighted strategy or, simply, about the justification for using the equally-weighted approach.

Regarding this point, we observe that behavioural finance represents a reference point because making decisions according to the “1 over N” rule can have a psychological basis.² First, this way of splitting the savings investment can be due to the belief of strong fallacies in the estimates of input parameters so that it is better to assume, a priori, that they are equal across asset classes and act consequently. Second, the equally-weighted portfolio can be a reasonable solution to avoid a feeling of regret that would occur if the investment decision consisted in a clear bet that comes out to be erroneous or suboptimal rather than in a mechanical and apparently neutral splitting. Finally, this naïve diversification can be the result of the search for variety when several choices have to be made simultaneously under uncertain conditions.

Example

Risk Budgeting

Equal Weight Approach

N=5



The Equally-weighted Approach

Table 4.1 Equally-weighted approach solutions for the numerical example

	Weights (%)	Marginal risk	Risk contribution	Percentage risk contribution (%)
A	20.00	-0.002948	-0.000590	-0.59
B	20.00	0.024125	0.004825	4.81
C	20.00	0.126998	0.025400	25.30
D	20.00	0.166741	0.033348	33.22
E	20.00	0.187061	0.037412	37.26

After providing these elements, the reason why the equally-weighted approach is included among risk-based strategies, albeit it works without any risk estimate, is because it is an attempt of defence toward risk (alternatively interpreted just as uncertainty or negative event), a sort of (unsophisticated) risk management tool. Certainly, the equally-weighted approach is motivated neither by a focus on a return target nor by the idea of being able to exploit particular performance-generating skills.

Nevertheless, the assumption that using this simple and heuristic method guarantees diversification can be misleading; it depends on the characteristics of asset classes that the strategy does not take into account. For instance, if their risk is extremely different, the equally-weighted scheme can lead to concentrated risk loadings.

The Equally-weighted Approach

It is clear that even if the portfolio is perfectly balanced in terms of capital allocation, its risk is mainly driven by asset classes C, D and E while asset class A plays a little hedging effect; asset class B is a minor risk contributor. The equally-weighted portfolio is much more volatile than the optimal risk parity portfolio: 10.04 % standard deviation versus 4.62 %. We observe that, in the particular case of the equally-weighted portfolio, the formula for the portfolio standard deviation (σ_p^{EW}) becomes:

$$\begin{aligned}\sigma_p^{EW} &= \sqrt{\sum_{i=1}^N \left(\frac{1}{N}\right)^2 \sigma_i^2 + 2 \sum_{j \neq i}^N \left(\frac{1}{N}\right) \left(\frac{1}{N}\right) \sigma_i \sigma_j \rho_{ij}} = \sqrt{\left(\frac{1}{N}\right)^2 \left(\sum_{i=1}^N \sigma_i^2 + 2 \sum_{j \neq i}^N \sigma_i \sigma_j \rho_{ij} \right)} \\ &= \left(\frac{1}{N}\right) \cdot \sqrt{\left(\sum_{i=1}^N \sigma_i^2 + 2 \sum_{j \neq i}^N \sigma_i \sigma_j \rho_{ij} \right)} = \left(\frac{1}{N}\right) \cdot \sqrt{\left(\sum_{i=1}^N \sigma_i^2 + \sum_{i=1}^N \sum_{j \neq i}^N \sigma_i \sigma_j \rho_{ij} \right)}\end{aligned}\tag{4.2}$$

The Equally-weighted Approach

Also the previous expression (3.14) for the total risk contributions ($TRC_{\sigma_i}^{EW.}$) can be rewritten in a slightly different form as it is shown here:

$$TRC_{\sigma_i}^{EW.} = \frac{1}{N} \cdot \frac{\sigma_i^2 \frac{1}{N} + \sum_{j \neq i}^N \frac{1}{N} \sigma_i \sigma_j \rho_{ij}}{\sigma_P^{EW.}} = \frac{\sigma_i^2 + \sum_{j \neq i}^N \sigma_i \sigma_j \rho_{ij}}{N^2 \sigma_P^{EW.}} \quad (4.3)$$

According to (4.3), the marginal risk, for example, of asset class A, could be obtained as follows:

$$TRC_{\sigma_A}^{EW.} = \frac{4\%^2 + (4\% \cdot 7\% \cdot 0.30) + (4\% \cdot 15\% \cdot (-0.20)) + (4\% \cdot 18\% \cdot (-0.10)) + (4\% \cdot 20\% \cdot (-0.25))}{5^2 \cdot 10.04\%} = -0.000590 \quad (4.4)$$

3.4 The Optimal Risk Parity Strategy

The truest version of risk parity strategy is optimal risk parity. With this expression, we refer to the risk-based strategy that, among practitioners, is popular simply with the term risk parity and in the case of academics is frequently called equally weighted risk contribution strategy or portfolio (abbreviated as ERC portfolio).

Optimal risk parity strategy is deeply rooted in finance literature and ideas about risk budgeting for investment processes. It is well known that risk budgeting recommends portfolio analysis in terms of risk contributions rather than in terms of portfolio weights. In the portfolio construction phase, this means that the starting point in finding the appropriate portfolio is specifying the risk contributions preferred by the investor or the asset manager. Then, the resulting portfolio has asset classes' weights and, as a consequence, a capital allocation driven by the targeted risk allocation. Thus, in a portfolio construction process consistent with the risk budgeting approach, risk contributions (and second moments estimates) are used as inputs and portfolio weights are the outputs.³

To give a better representation of the portfolio construction methodology in the risk budgeting perspective, we consider a supposed set of risk budgets ($BD_1, \dots, BD_i, \dots, BD_N$)⁴ and the specification of the desired risk contributions ($TRC_1, \dots, TRC_i, \dots, TRC_N$) to a general risk measure RM .⁵ The problem of finding out the set of weights ($w_1, \dots, w_i, \dots, w_N$) such that the risk contributions match the preferred risk budgets takes the form of the following non linear mathematical system:⁶

³We note that in the classic *Mean-Variance Optimization* the portfolio weights are the main outputs and risk contributions are a secondary output determined by the first.

⁴Risk budgets (BD_i) are indicated in relative terms (that is they are expressed in %), not as nominal values.

⁵From Sect. 3.2, we remember that a general TRC_i corresponds to $w_i \cdot \frac{\partial RM}{\partial w_i}$.

⁶In spite of the general form of the mathematical system we have adopted, we observe that with reference to the constraint on the risk budgets, Roncalli (2014) considers not very reasonable to set one risk budget to zero, it would be better to reduce the investment universe excluding the corresponding asset in that case. For this reason, he suggests a strict positivity constraint on the BD_i .

$$\begin{cases}
 w_1 \cdot \frac{\partial RM}{\partial w_1} = BD_1 \cdot RM \\
 \vdots \\
 w_i \cdot \frac{\partial RM}{\partial w_i} = BD_i \cdot RM \\
 \vdots \\
 w_N \cdot \frac{\partial RM}{\partial w_N} = BD_N \cdot RM \\
 \\
 BD_1, \dots, BD_i, \dots, BD_N \geq 0 \\
 BD_1 + \dots + BD_i + \dots + BD_N = 1 \\
 w_1, \dots, w_i, \dots, w_N \geq 0 \\
 w_1 + \dots + w_i + \dots + w_N = 1
 \end{cases} \quad (3.27)$$

However, regarding optimal risk parity, it is not possible to analytically solve a mathematical system like the one in (3.27). For this reason, we will subsequently focus on a different way to formalize the same problem and will give an intuitive explanation as how solutions are identified.

Before we look at these technical aspects, we illustrate qualitatively the optimal risk parity. The main idea inspiring this strategy is that of preventing one or few asset classes from having a dominant role in driving portfolio risk, typically measured by the standard deviation. Consequently, the strategy seeks to equalize risk contributions from the different asset classes in the investment universe at least on an ex-ante basis. In other words, the optimal risk parity strategy wants the portfolio to be equally weighted or perfectly balanced in terms of risk allocations.

Regarding the technical issues, it should be clear from the above description that implementing optimal risk parity strategy is equivalent to finding out a risk budgeting portfolio with uniform (and obviously strictly positive) risk budgets assigned to each asset class. To formalize the concepts just expressed, the mathematical system in (3.27) must be rewritten as shown in (3.28):

$$\begin{cases}
 w_i \cdot \frac{\partial \sigma_P}{\partial w_i} = BD \cdot \sigma_P \\
 BD = 1/N \\
 \sum_{i=1}^N w_i = 1 \\
 w_i \geq 0
 \end{cases} \quad (3.28)$$

As mentioned before, it is difficult to find explicitly the solutions w_i for the risk parity portfolio in a general context where standard deviations and correlations are

not homogeneous among portfolio constituents.⁷ Maillard et al. (2010) have provided some intuitions on the appropriate weights for the ERC portfolio but again they do not lead to closed-form solutions. In particular, they first recall the definition of β_i as follows:

$$\beta_i = \frac{\sum_{j=1}^N w_j \sigma_{ij}}{\sigma_P^2} \quad (3.29)$$

Second, they link this definition with the one concerning risk contribution:

$$TRC_{\sigma_i} = w_i \beta_i \sigma_P \quad (3.30)$$

Considering the condition of homogeneous risk contributions for the optimal risk parity strategy equivalent to σ_P/N , they conclude that:

$$w_i = \frac{\beta_i^{-1}}{\sum_{j=1}^N \beta_j^{-1}} = \frac{\beta_i^{-1}}{N} = \frac{1}{N \beta_i} \quad (3.31)$$

According to (3.31), the weight of asset i belonging to a risk parity portfolio should be inversely proportional to its beta: the higher the beta, the lower the weight and vice versa. Obviously, this also implies that an asset class with high individual volatility and/or with high correlation with the other asset classes becomes penalized in the portfolio allocations. Despite the interesting insight, Eq. (3.31) doesn't make the problem of finding the portfolio weights for the optimal risk parity strategy analytically tractable. The reason is that asset allocation w_i is function of itself or, said differently, the beta cannot be found without the portfolio weights, but they have to depend on beta. The problem that arises is referred by Chaves et al. (2012), Maillard et al. (2010) and Roncalli (2014) as an endogeneity problem.

Taking into account both the problem just mentioned and the impossibility to get a closed-form solution for the system in (3.28), it makes sense to search differently the weights for an ERC portfolio.

A suitable approach requires first to rewrite the non-linear system in (3.28) as a constrained optimization problem. Its form is given in (3.32):

$$\begin{aligned} w^* &= \arg \min f(w) \\ \text{s.t. } \sum_{i=1}^N w_i &= 1 \\ 0 &\leq w_i \leq 1 \end{aligned} \quad (3.32)$$

⁷Explicit solutions are admissible just in the bivariate case and in the case of constant correlation. See Sect. 3.3.

Secondly, a reasonable objective function $f(w)$ to minimise is required. For this purpose, it is necessary to remark the condition that must be satisfied by each asset class allocation in an optimal risk parity portfolio. Considering that the main goal of the strategy is to distribute the same risk budget/contribution to each component, so that none has a dominant role on the portfolio risk, the condition can be formally translated as follows:

$$w_i \cdot \frac{\partial \sigma_P}{\partial w_i} = w_j \cdot \frac{\partial \sigma_P}{\partial w_j} \quad \forall i, j \quad (3.33)$$

Accordingly, the objective function to be minimised is then given by the following mathematical expression:

$$f(w) = \sum_{i=1}^N \sum_{j=1}^N \left(w_i \cdot \frac{\partial \sigma_P}{\partial w_i} - w_j \cdot \frac{\partial \sigma_P}{\partial w_j} \right)^2 \quad (3.34)$$

Now, the optimization problem to solve for w^* (the optimal risk parity portfolio weights) can be fully stated as follows:

$$\begin{aligned} & \underset{w^*}{\text{Min}} \sum_{i=1}^N \sum_{j=1}^N \left(w_i \cdot \frac{\partial \sigma_P}{\partial w_i} - w_j \cdot \frac{\partial \sigma_P}{\partial w_j} \right)^2 \\ & \text{s.t.} \\ & \sum_{i=1}^N w_i = 1 \\ & 0 \leq w_i \leq 1 \end{aligned} \quad (3.35)$$

We note that (3.35) is a problem of constrained nonlinear programming for which analytical solutions are not available and thus we must find them using a numerical algorithm that implements an iterative process. Maillard et al. (2010) and Roncalli (2014) recommend the use of the Sequential Quadratic Programming Algorithm (SQP).

This method generates approximated solutions that allow convergence to the minimum of the nonlinear optimization problem. So, SQP produces a sequence of solutions that get closer and closer to the right one w^* . The tentative solutions are obtained using the current iterate simultaneously to replace the original objective function with a quadratic problem approximation and to approximate the constraint functions after linearizing them.⁸

⁸It is useful to keep in mind that try to model a constrained nonlinear optimization problem as a quadratic programming subproblem means write the second-order Taylor expansion of $f(w)$. Since it is an N -dimensional function, the expansion involves the gradient and the hessian of the function. In particular, the quadratic subproblem at the current approximation w_s has the form:

We observe that, besides the Sequential Quadratic Programming Algorithm, other authors (Chaves et al. 2012) have proposed different approaches to compute the appropriate allocations for an ERC portfolio presenting them as an important simplification for implementing the optimal risk parity strategy. It has to be noted they are based on matrix algebra; neither do they give a specification of an objective function nor do they allow for the insertion of weight constraints.

So far we have described the theoretical concepts and the tools on which the optimal risk parity rests. Next we apply this strategy for illustrative purpose to an investment universe of five risky asset classes denoted A, B, C, D and E. Their volatilities are respectively 4, 7, 15, 18 and 20 %. The correlation coefficients between couples are provided in Table 3.4 and Table 3.5 provides the corresponding covariance matrix.

We compute the weights for the optimal risk parity portfolio using the Sequential Quadratic Programming Algorithm, therefore we follow Maillard et al. (2010) and Roncalli (2014). The solutions are given in Table 3.6 and lead to 4.62 % portfolio standard deviation. Unsurprisingly, the optimal risk parity portfolio is invested in all asset classes. We observe substantial differences in the assigned wealth allocation. In particular, we note a significant exposure in asset class A and B, the ones with both the lower standard deviation and the lower correlation with C, D and F. Importantly, Table 3.6 gives evidence that the strategy implementation has determined the decomposition of the portfolio standard deviation into perfectly balanced separate components. Indeed, no asset class contributes more than its

Table 3.4 Correlation matrix for the numerical example

	A	B	C	D	E
A	1.00	0.30	-0.20	-0.10	-0.25
B	0.30	1.00	0.10	0.20	0.20
C	-0.20	0.10	1.00	0.70	0.75
D	-0.10	0.20	0.70	1.00	0.85
E	-0.25	0.20	0.75	0.85	1.00

(Footnote 8 continued)

$$\text{Min}_{\mathbf{d}_w} \nabla f(\mathbf{w}_s) \mathbf{d}_w + \frac{1}{2} \mathbf{d}_w' \nabla^2 f(\mathbf{w}_s) \mathbf{d}_w$$

s.t.

$$\nabla h_f(\mathbf{w}_s) \mathbf{d}_w + h_f(\mathbf{w}_s) = 0 \text{ with } f = 1, \dots, F$$

$$\nabla y_g(\mathbf{w}_s) \mathbf{d}_w + y_g(\mathbf{w}_s) \geq 0 \text{ with } g = 1, \dots, G$$

where $\mathbf{d}_w = \mathbf{w} - \mathbf{w}_s$.

The solution of the subproblem is used as a search direction to determine the next iterate. The final solution must satisfy the so called Karush-Kuhn-Tucker conditions that, in this case, are referred to a Lagrangian function that takes into account the constraints.

Table 3.5 Covariance matrix for the numerical example

	A	B	C	D	E
A	0.001600	0.000840	−0.001200	−0.000720	−0.002000
B	0.000840	0.004900	0.001050	0.002520	0.002800
C	−0.001200	0.001050	0.022500	0.018900	0.022500
D	−0.000720	0.002520	0.018900	0.032400	0.030600
E	−0.002000	0.002800	0.022500	0.030600	0.040000

Table 3.6 Optimal risk parity solutions for the numerical example

	Weights (%)	Marginal risk	Risk contribution	Percentage risk contribution (%)
A	55.41	0.016680	0.009241	20.00
B	21.55	0.042878	0.009241	20.00
C	9.49	0.097344	0.009241	20.00
D	6.88	0.134346	0.009241	20.00
E	6.67	0.138587	0.009241	20.00

peers (or competitors) to the overall volatility. A 20 % portion of it (that is 0.009241) can be assigned to each individual portfolio constituent.

For the sake of clarity, it is helpful to highlight the critical role played by correlations in our illustrative example. To this end, we simply compute the portfolio weights using the naïve risk parity, the simplified version of the strategy that ignores correlations.

It is immediately clear how different the portfolio would be. As a matter of fact, we would get as result the following allocations vector: 44.24, 25.28, 11.80, 9.83 and 8.85 %. Even if they are ranked in the same order of magnitude, the discrepancies with the column “Weights” in Table 3.6 are somewhat substantial. The naïve strategy has a larger volatility than the optimal risk parity strategy (5.62 % vs. 4.62 %) and from Table 3.7 we can easily note that it displays imbalances in terms of risk contribution. In particular, asset class A has the lowest share in the portfolio volatility which is intuitive given that its low correlations with the other asset classes are not considered in the decision on portfolio weights.

Table 3.7 Naïve risk parity solutions for the numerical example

	Weights (%)	Marginal risk	Risk contribution	Percentage risk contribution (%)
A	44.24	0.009440	0.004176	7.43
B	25.28	0.039647	0.010023	17.82
C	11.80	0.110917	0.013086	23.27
D	9.83	0.150092	0.014756	26.23
E	8.85	0.160476	0.014199	25.25